IIT ROORKEE

NPTEL

NPTEL ONLINE
CERTIFICATION COURSE

REFRIGERATION AND AIR-CONDITIONING

RECPULATION OF THERMODYNAMICS

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MECHANICAL & INDUSTRIAL ENGINEERING

Topics to be covered

- Definition of thermodynamics
- Basic terms in thermodynamics
- Laws of thermodynamics
- Thermodynamic processes

Thermodynamics

The science that encompasses the study of energy, its transformation and relationship amongst the various physical quantities of substance which are affected by or cause these transformations.

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DEFINITIONS

System

Prescribed region of space or finite quantity of matter surrounded by an envelope called the boundary.

Closed System, open system and isolated system

Surrounding

The space and matter external to the thermodynamic system and outside boundary is called the surrounding.

Universe

When system and the surrounding are put together it is called universe.

Work

Work is done by a system if the sole effect on the surroundings (everything external to the system) could be raising of a weight.

Heat

Heat is defined as the form of energy that is transferred across the boundary of a system at a given temperature to another system (or the surrounding) at a lower temperature by virtue of temperature difference between the two systems.

Property

Measurable characteristics describing the system that depends on the state of the system and is independent of the path (that is, the prior history) by which the system arrived at the given state.

- Intensive Property
- Extensive Property

State

Unique condition of the system, at an instant of time, described by its properties.

Process

Path of succession of state points through which the system passes during transition from one state to another state is called a process.

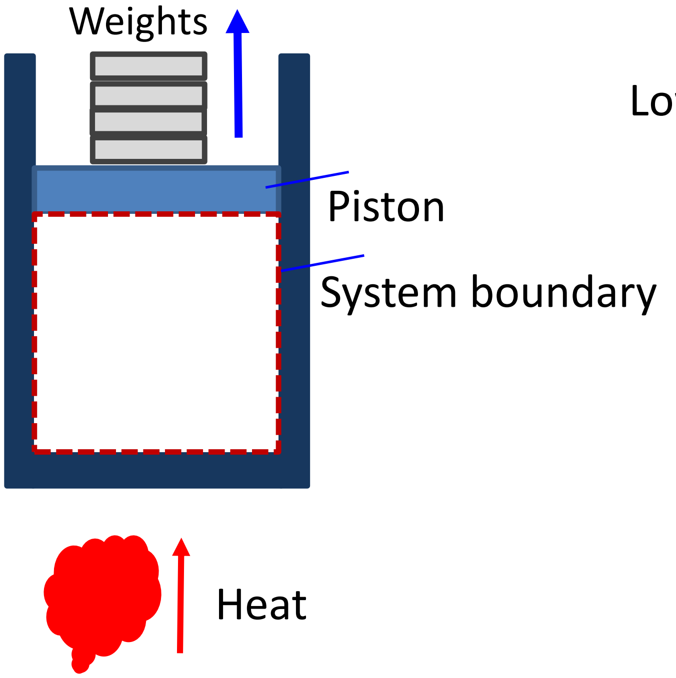
Cycle

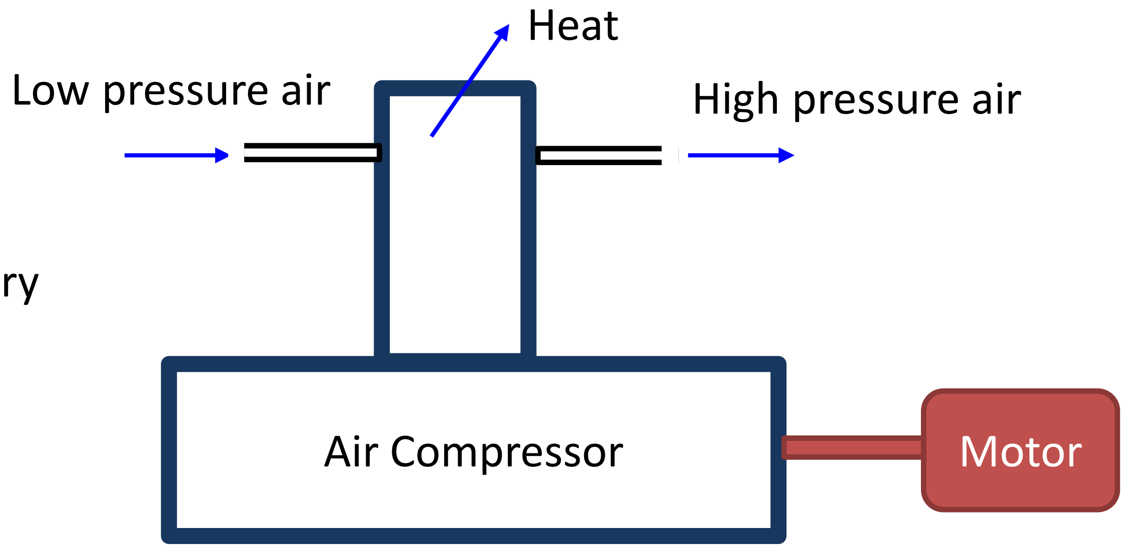
When a system in a given initial state goes through a number of different changes of states or processes and finally returns to the initial state, the system has undergone a cycle.

Work

- Boundary work $\int pdv$
- Shaft work or flow work $-\int vdp$

Control Mass and Control Volume





Boundary work $\int pdv$

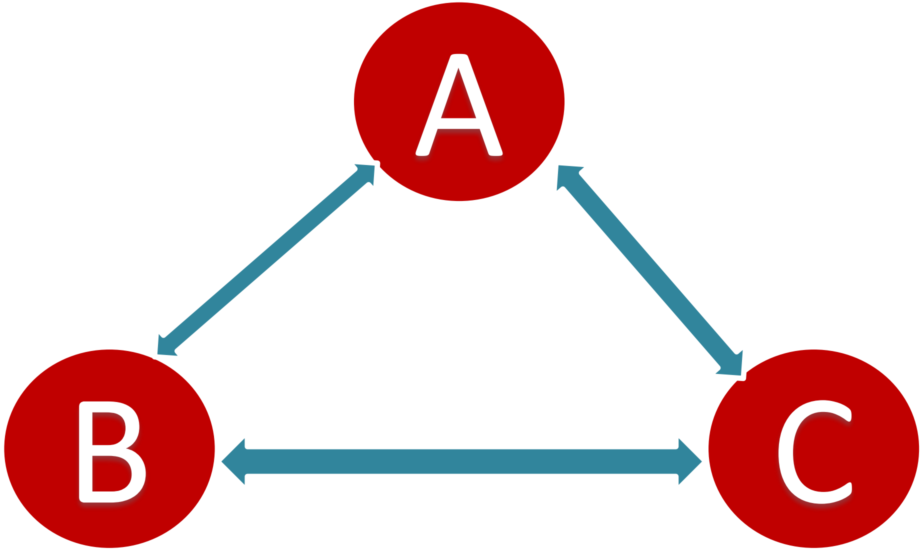
Shaft work $-\int vdp$

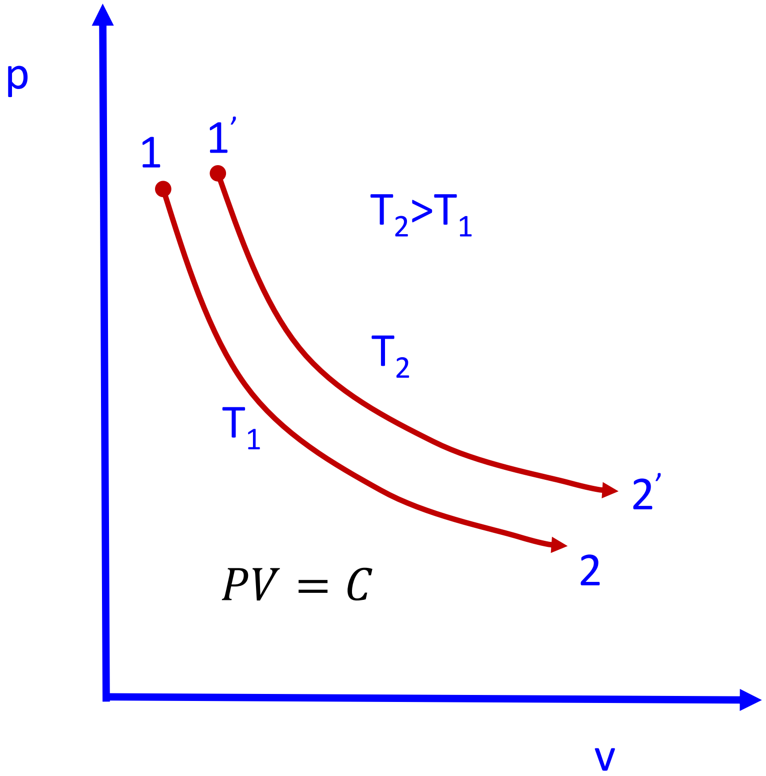
Laws of Thermodynamics

- Zeroth Law
- First Law
- Second Law
- Third Law

Zeroth Law

The zeroth law of thermodynamics states that if two thermodynamic systems are each in thermal equilibrium with a third, then they are in thermal equilibrium with each other.




p
1
1'
T2>T1
T2
T1
2'
2
PV = C
v

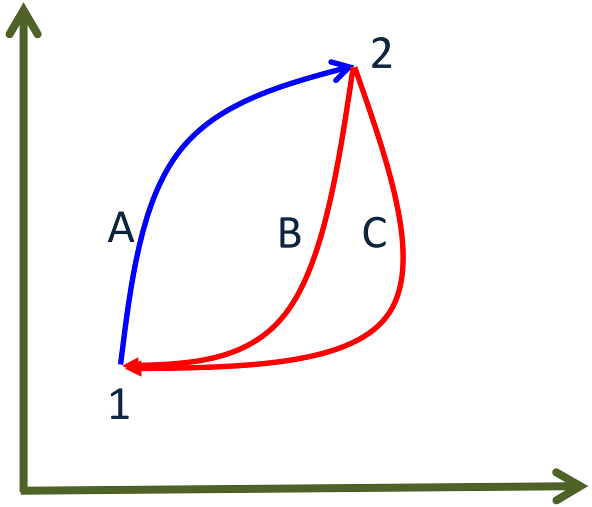
First Law of Thermodynamics

In a cyclic process the cyclic integral of heat transfer to the system is equal to the cyclic integral of the work transfer to the surrounding.

$$\oint \delta Q = \oint \delta W$$

$$\int_1^2 \delta Q_A + \int_2^1 \delta Q_B = \int_1^2 \delta W_A + \int_2^1 \delta W_B$$

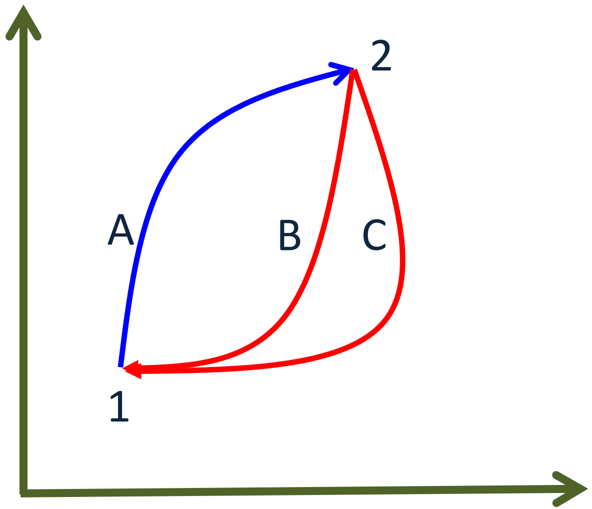
$$\int_1^2 \delta Q_A + \int_2^1 \delta Q_C = \int_1^2 \delta W_A + \int_2^1 \delta W_C$$



$$\int_1^2 \cancel{\delta Q_A} + \int_2^1 \delta Q_B = \int_1^2 \cancel{\delta W_A} + \int_2^1 \delta W_B$$

$$\int_1^2 \cancel{\delta Q_A} + \int_2^1 \delta Q_C = \int_1^2 \cancel{\delta W_A} + \int_2^1 \delta W_C$$

$$\int_2^1 \delta Q_B - \int_2^1 \delta Q_C = \int_2^1 \delta W_B - \int_2^1 \delta W_C$$



$$\int_2^1 \delta Q_B - \int_2^1 \delta W_B = \int_2^1 \delta Q_C - \int_2^1 \delta W_C$$

$$\int (\delta Q - \delta W)_B = \int (\delta Q - \delta W)_C$$

$$\delta Q - \delta W = dE$$

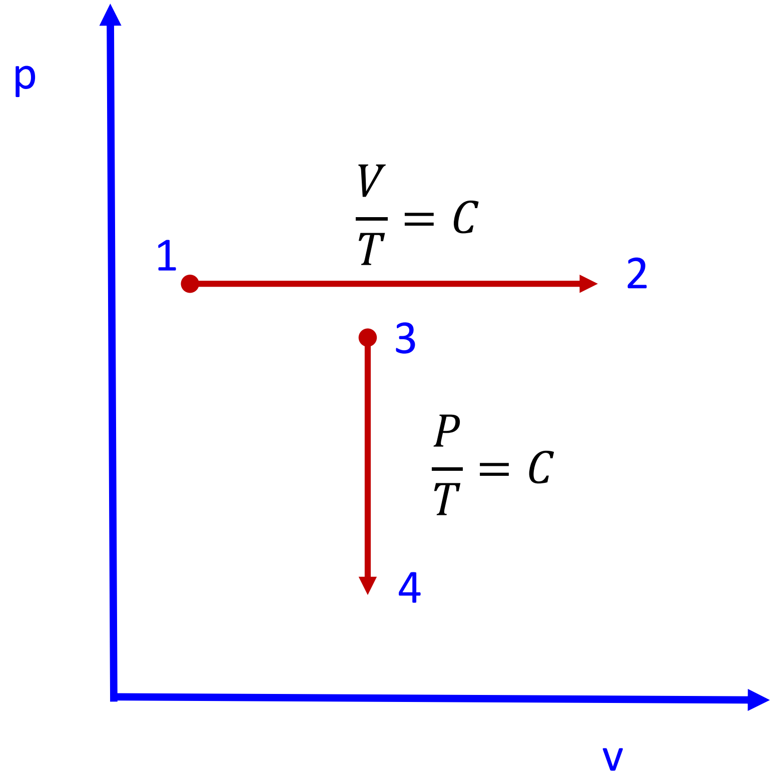
$$\delta Q_{1-2} - \delta W_{1-2} = \Delta KE_{1-2} + \Delta PE_{1-2} + U_2 - U_1$$

$$\delta \mathrm{q} - \delta \mathrm{w} = \mathrm{u}_2 - \mathrm{u}_1$$

- The internal energy of a closed system remains unchanged if the system is isolated from the surroundings.
- A perpetual motion of machine of first kind is impossible.

Thermodynamics process for closed system

- Isothermal process
- Isobaric process
- Isochroic process
- Polytropic process
- Adiabatic process
- Isentropic process


p
1
2
3
4
$\frac{V}{T} = C$
$\frac{P}{T} = C$
v

First Law of Thermodynamics for Open System

$$\delta Q - \delta W = dh + CdC + gdZ$$

$$Q - W = (h_2 - h_1) + \frac{C_2^2 - C_1^2}{2} + g(Z_2 - Z_1)$$

- Turbine
- Compressor
- Nozzles
- Heat Exchanger

Second Law of Tehrmodynamics

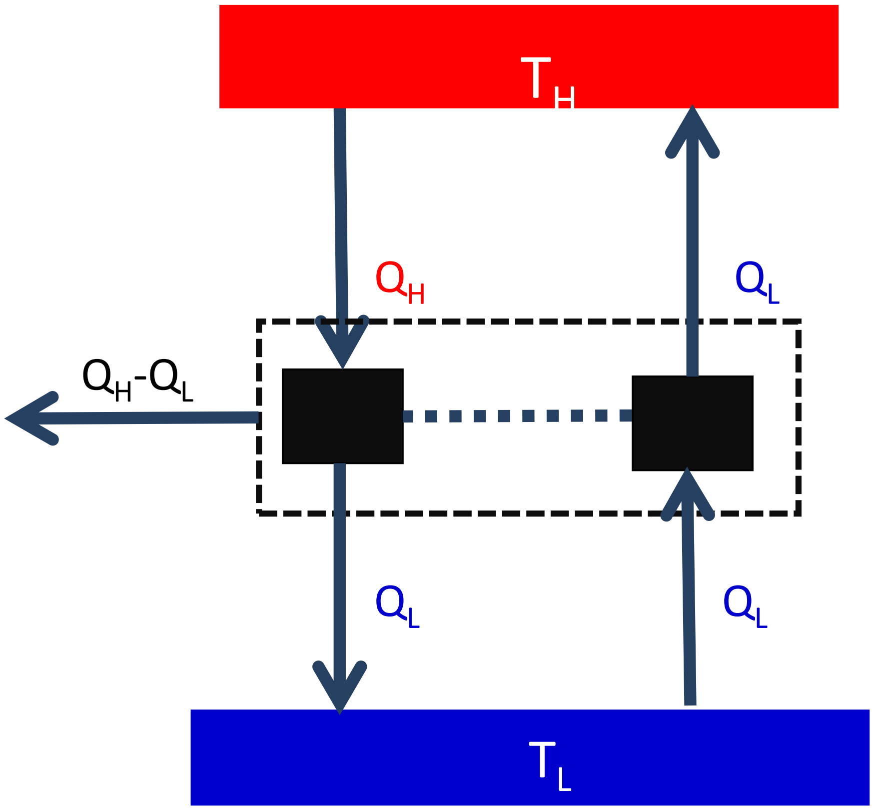
Kelvin-Plank Statement

It is impossible to construct a device that will operate in a cycle and produces no effect other than the rising of a weight and the exchange of heat with a single reservoir.

Clausius Statement

It is impossible to construct a device that operates in a cycle and produces no effect other than the transfer of heat from a cooler body to a hotter body.

Assume violation of Clausius statement heat pump can be eliminated.



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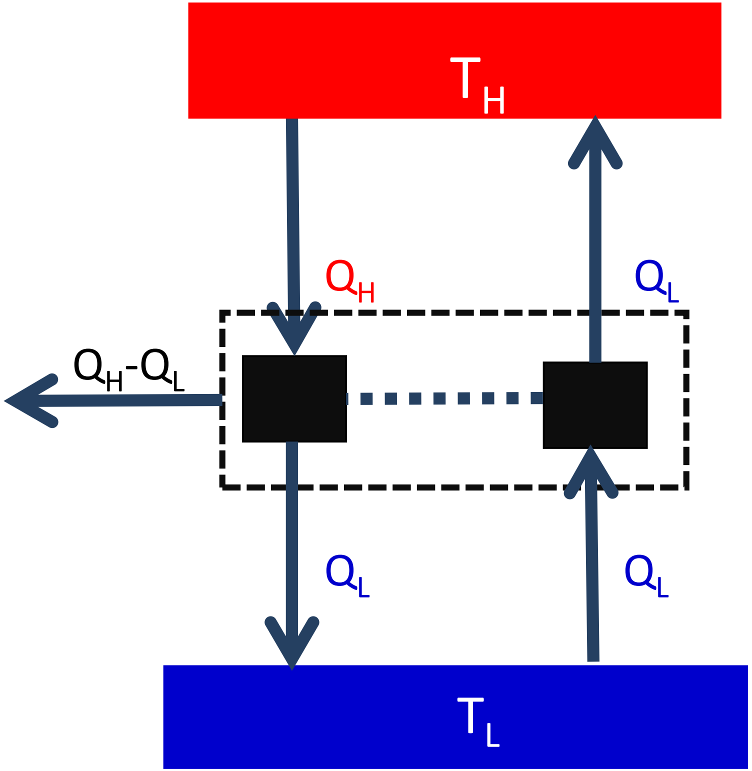
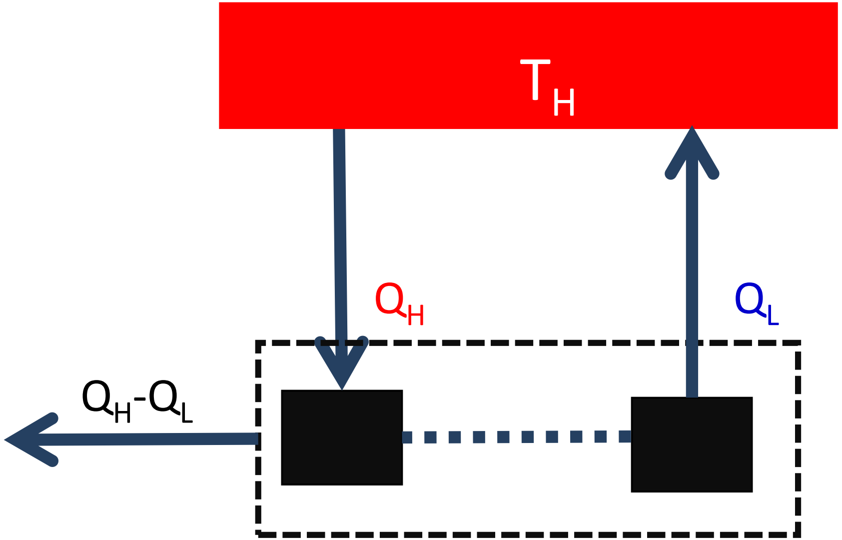
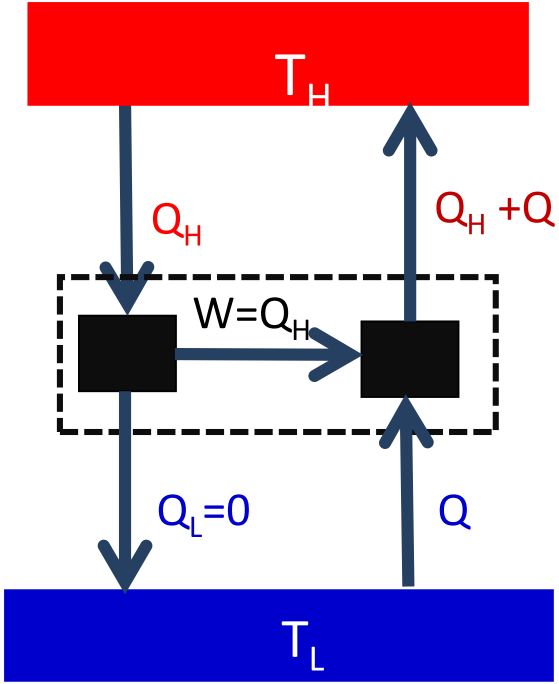


T_H
Q_H
Q_L
Q_H-Q_L
Q_L
Q_L
T_L

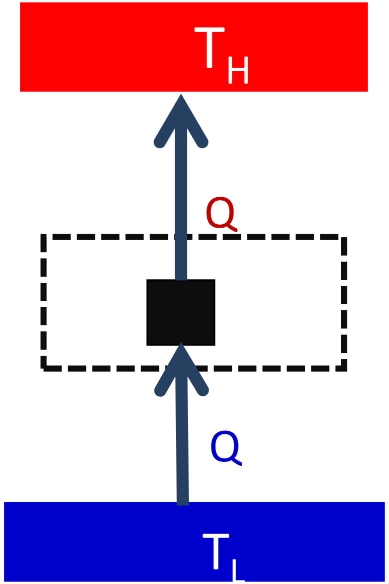


T_H
Q_H
Q_L
Q_H-Q_L

Violation of Plank's statement



Violation of Clausius statement

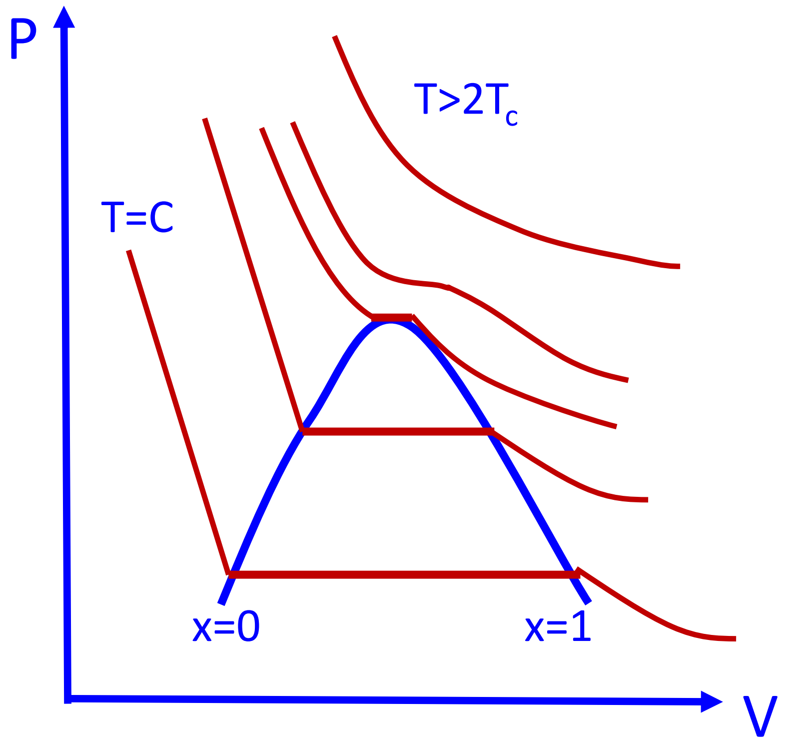


Gases and Vapours

- The vapour is a gas when the temperature is greater than twice the critical temperature.
- Critical Temperature of water

  $= 373.95\ ^oC = 373.95+273.15 = 647.1\ K$
- Water can be considered as gas at $2x647.1 = 1294.2\ K$

  $1294.2 - 273.15 = 1021.05\ ^oC.$
- Any vapour below 10 kPa pressure can be considered as a gas.



- Therefore, in air-conditioning the water vapour can be considered as the ideal gas not in the case of steam power plant.
- For steam power plant p>10 MPa and $T_{max} \approx 640\ ^oC$.
- Hence, ideal gas equation does not holds good.

The End